

The Lifecycle of a Listed Future

The Ledger — companion to `FutureLifeCycle.hs`

Abstract

A listed future is recorded, traded, settled daily for variation margin, expired, and flattened to zero within the three-map state model of the StatesHome addendum (§4.1) over the move and conservation primitives of v10.3. Each step is one atomic `StateDelta`; conservation is shown at each, not asserted. The variation-margin settlement is the centrepiece: one event that writes the shared settlement price once and fans out over the current holders, resetting each accumulated cost and moving cash. One worked example — one cash-settled future, multiplier 50, wallets A, B, C — runs the whole life through the terminal `Close`, which flattens every position to $(0, 0, 0)$. The reference implementation is `FutureLifeCycle.hs`, whose `main` computes these figures through `Close`; this document reproduces them exactly in prose and writes no Haskell.

1 Scope and primitives recalled

The framework primitives are those of v10.3 and are used here in silence: a *wallet* w holds signed quantities $h(w, u)$ of *units* u ; a *move* transfers a positive quantity of one unit from a source wallet to a destination wallet, with semantics $w_s(u) \text{--} = q$, $w_d(u) \text{+} = q$; a *transaction* is a finite set of simultaneous moves; *conservation* is the invariant $\sum_w h(w, u) = 0$ for every u , an algebraic consequence of the move pattern, not a runtime check. Write $\text{net_qty}(w, u) := h(w, u)$ for the signed quantity of unit u held by wallet w ; the formulas below use `net_qty`.

State attaches to three objects, each with its own map and mutation discipline (addendum §1):

Map	Key	Discipline
<code>ProductTerms</code>	u	immutable, append-only, versioned; registration-total
<code>UnitStatus</code>	u	mutable, shared across holders; registration-total
<code>PositionState</code>	(w, u)	monotone carrier; <code>Option</code> accessor

For the future, the fields divide by who reads them and how often they change:

Field	Home	Reason
<code>multiplier</code> , <code>currency</code> , <code>expiry</code> , <code>clearinghouse</code> , <code>exchange</code> , <code>product_id</code>	<code>ProductTerms[u_F]</code>	immutable; the CME and ICE contracts are distinct units.
<code>lifecycle_stage</code> , <code>last_settlement_price</code> , <code>last_settlement_date</code>	<code>UnitStatus[u_F]</code>	one settlement price per contract, read identically by every holder.
<code>accumulated_cost</code> (<code>ac</code>)	<code>PositionState[w, u_F]</code>	two wallets holding the same contract carry different <code>ac</code> .

The accumulated cost `ac` is a *conserved* field: $\sum_w \text{ac}(w, u) = 0$ for every u , enforced one level up at the event handler (addendum C2), since a refinement type on a sum of exact integers is free in no production language.

The atomic delta. One event proposes one **StateDelta** for one unit across the three maps: at most one shared **Stage** write, a map of per-holder position deltas ($\Delta\text{net_qty}, \Delta\text{ac}$), and a map of per-holder cash legs. It applies in full or not at all (C3); a partially applied delta is not representable. A delta reaches application only through **validate**, which discharges conservation — $\sum_w \Delta f(w, u) = 0$ for each conserved field — and returns the abstract witness **ValidDelta**. An unconserved delta cannot be applied.

The single dimension bridge. Contracts, money, and price are distinct types (**Qty**, **Cash**, **Price**). The only multiplication in the lifecycle,

$$\text{markValue}(\text{net_qty}, S, m) = \text{net_qty} \cdot S \cdot m : \text{Cash},$$

converts contracts at a price into money. **Price** carries no addition: two prices are never summed, nor moved between wallets. This makes the load-bearing identity $\text{VM} = \text{net_qty} \cdot S \cdot m + \text{ac}$ typecheck only because both summands are **Cash**. **Price** and **Cash** share one fixed minor-unit scale, chosen so that $\text{net_qty} \cdot S \cdot m$ lands in **Cash**'s unit. The worked example carries **Price** in whole index points and **Cash** in whole dollars for readability; a production scale carries **Price** in scaled minor units — a quarter-point for ES, a cent for a bond quoted in 32nds — so sub-point ticks are exact. The arithmetic is integer throughout and never divides, so the scale is free to fix and no rounding can break conservation at any scale.

2 The instrument and the worked example

One cash-settled listed future $u_F = \text{ES-FUT}$, with

$m = 50$, **currency** = USD, **expiry** = Day 3, **clearinghouse** = CME-CH, **exchange** = CME.

Wallets A, B, C are traders; CH is the clearinghouse virtual wallet — the variation-margin settlement hub, whose cash leg is the residual that balances the holders' legs. CH stands in for the clearinghouse as a settlement hub, not as a *novating* central counterparty: faithful novation would interpose CH on the *contract* leg of every trade ($A+q / \text{CH}-q$ and $\text{CH}+q / B-q$), so the two cleared sides never face each other. The clearing infrastructure itself is out of scope; here CH carries only the variation-margin cash residual. Two rules govern every figure below:

- **Trade.** For a signed-quantity change Δ at price p , each leg sets $\text{ac} += -\Delta \cdot p \cdot m$.
- **Settlement.** At price S , the per-holder variation margin is $\text{VM}(w) = \text{net_qty}(w) \cdot S \cdot m + \text{ac}(w)$ (cash, routed through CH); then $\text{ac}(w) \leftarrow -\text{net_qty}(w) \cdot S \cdot m$.

The whole life, with **net_qty**, **ac**, and the per-step variation-margin cash for (A, B, C) :

Step	Event	net_qty (A,B,C)	ac (A,B,C)	VM c
Listing	u_F registered; stage REGISTERED , no mark; no rows	(0, 0, 0)	(−, −, −)	
T1	A buys 10 from B @100	(10, −10, 0)	(−50000, +50000, 0)	
Settle d1	$S_1 = 102$; shared price $\leftarrow 102$	(10, −10, 0)	(−51000, +51000, 0)	(+100, 0, 0)
T2	C buys 4 from A @103	(6, −10, 4)	(−30400, +51000, −20600)	
Settle d2	$S_2 = 101$; shared price $\leftarrow 101$	(6, −10, 4)	(−30300, +50500, −20200)	(−100, 0, 0)
T3	B buys 4 from C @101 (C $\rightarrow$ flat)	(6, −6, 0)	(−30300, +30300, 0)	
Expiry	stage $\rightarrow$ EXPIRED ; final $S = 105$	(6, −6, 0)	(−31500, +31500, 0)	(+120, 0, 0)
Close	units returned to CH; positions $\rightarrow 0$	(0, 0, 0)	(0, 0, 0)	

The sections below walk this table in the order the events occur. Each event is one **StateDelta**; each shows the three conservation sums $\sum \Delta\text{net_qty}$, $\sum \Delta\text{ac}$, $\sum \text{VM}$. The example is the argument; the prose names the mechanism.

3 Listing

Registration writes $ProductTerms[u_F]$ (the six immutable terms above) and $UnitStatus[u_F]$ together, establishing “ $u \in \text{terms} \iff u \in \text{status}$ ” by construction (**register**). The status is initialised to stage **REGISTERED** with no settlement mark. No $PositionState$ row exists: $position(w, u_F) = \text{None}$ for every w .

REGISTERED denotes a unit recorded in and known to the ledger before any settlement, not admission to an exchange; an OTC option or an internal strategy reaches it identically. The two unreachable status states — a never-traded unit carrying a price, an expired unit lacking a final mark — are made unspellable by fusing the mark onto the stage:

$$\text{Stage} = \text{Registered} \mid \text{Active}(\text{Maybe Settlement}) \mid \text{Expired Settlement}.$$

Registered carries no mark; **Expired** always carries one. The addendum’s `last_settlement_price` and `last_settlement_date` are projections of the **Settlement** carried by the stage, not independent fields. Conservation holds vacuously: no holders, so every sum is the empty sum, zero (C9). Re-registering an existing unit id is a hard error, never a silent reset (C10).

4 First trade

A buys 10 contracts from B at 100. The trade emits one **StateDelta**:

- **Shared.** **REGISTERED** $\rightarrow$ **ACTIVE** with no mark (**Active Nothing**); the first trade activates the unit. Any existing settlement mark would be preserved — a trade never wipes the shared price.
- **Per-position.** The contract quantity moves from seller to buyer: $\Delta \text{net_qty}(A) = +10$, $\Delta \text{net_qty}(B) = -10$. Each leg sets $\text{ac} += -\Delta \cdot p \cdot m$:

$$\Delta \text{ac}(A) = -(+10) \cdot 100 \cdot 50 = -50000, \quad \Delta \text{ac}(B) = -(-10) \cdot 100 \cdot 50 = +50000.$$

- **Cash.** None: entering a futures position pays no notional.

Conservation: $\sum \Delta \text{net_qty} = +10 - 10 = 0$, $\sum \Delta \text{ac} = -50000 + 50000 = 0$, $\sum \text{VM} = 0$. State after T1: $\text{net_qty} = (10, -10, 0)$, $\text{ac} = (-50000, +50000, 0)$. A and B cross from None to Some on first touch; C is still None.

Positive quantity. A trade leg carries a strictly positive quantity $q > 0$, parsed at the boundary (`mkPosQty`); $q \leq 0$ is rejected (`NonPositiveQty`) and never recorded. $q = 0$ would promote **REGISTERED** $\rightarrow$ **ACTIVE** and create Some-flat rows for two never-held wallets, collapsing the never-held/held-flat distinction (§8); $q < 0$ would swap the buyer and seller roles, violating the move primitive’s positivity. The sign of the position change is supplied by the leg — buyer $+q$, seller $-q$ — not by q .

5 Daily variation-margin settlement: the mechanism and the clean case (day 1)

Settlement at price S , multiplier m , over the current holders. For each holder the reset target is $\text{target} = -\text{net_qty} \cdot S \cdot m$, the row delta is $\Delta \text{ac} = \text{target} - \text{ac}$, and the cash leg is

$$\boxed{\text{VM}(w) = -\Delta \text{ac}(w) = \text{net_qty}(w) \cdot S \cdot m + \text{ac}(w)}$$

This is the centrepiece identity. The cash leg is the mirror of the change in accumulated cost. Therefore $\sum_w \text{VM} = -\sum_w \Delta \text{ac}$, and variation-margin zero-sum is the *same* fact as *ac* conservation, not a separate reconciliation.

Settlement is therefore one atomic **StateDelta** touching both layers:

- **Shared**, one write: $\text{UnitStatus}[u_F]$ acquires the mark (S, d) . The *coarse stage rank* — **REGISTERED** < **ACTIVE** < **EXPIRED** — is unchanged by a settle: it stays **ACTIVE**. What updates every settle is the *embedded mark*, the stage value becoming **Active (Just (Settlement S d))**.
- **Per-position**, a fan-out over the current holders: each row's *ac* is reset to target, and each holder gets the cash leg $\text{VM}(w)$.

The cash legs are emitted over the holders. The clearinghouse leg is the residual that balances them; since the holder legs already sum to zero, CH's leg is zero throughout this example and no CH cash row materialises. This is the special case of a *closed* holder set $\{A, B, C\}$, in which $\sum_w \text{net_qty}(w) = 0$ at every step. In the Ledger's intended use — an internal firm system of record facing an external clearinghouse — the firm is generally *not* flat: when $\sum_w \text{net_qty}(w) \neq 0$ over the firm's holders, the CH leg is the firm's nonzero net variation margin, and that residual is exactly the boundary figure reconciled against the clearing broker's daily statement.

Day 1: $S_1 = 102$. The holders are A and B; C has no row and is untouched — the contract settles vacuously over C, the shared price still updating (C9). With pre-settle *ac* = (−50000, +50000) on (A, B) and *net_qty* = (10, −10):

$$\text{VM}(A) = 10 \cdot 102 \cdot 50 + (-50000) = 51000 - 50000 = +1000,$$

$$\text{VM}(B) = -10 \cdot 102 \cdot 50 + 50000 = -51000 + 50000 = -1000.$$

The *ac* reset: $\text{ac}(A) \leftarrow -10 \cdot 102 \cdot 50 = -51000$, $\text{ac}(B) \leftarrow +51000$. Conservation: $\sum \Delta \text{net_qty} = 0$ (no quantity moved), $\sum \Delta \text{ac} = (-51000 + 50000) + (51000 - 50000) = 0$, $\sum \text{VM} = +1000 - 1000 = 0$. State: *net_qty* = (10, −10, 0), *ac* = (−51000, +51000, 0).

6 A subsequent, intraday trade (day 2)

C buys 4 contracts from A at 103, after the day-1 settle and before the day-2 settle — A trades intraday. One **StateDelta**, stage already **Active** (mark preserved):

$$\Delta \text{net_qty}(C) = +4, \quad \Delta \text{net_qty}(A) = -4; \quad \Delta \text{ac}(C) = -(+4) \cdot 103 \cdot 50 = -20600, \quad \Delta \text{ac}(A) = +20600.$$

Conservation: $\sum \Delta \text{net_qty} = +4 - 4 = 0$, $\sum \Delta \text{ac} = -20600 + 20600 = 0$, $\sum \text{VM} = 0$ (a trade moves no variation-margin cash). State after T2: *net_qty* = (6, −10, 4), *ac* = (−51000 + 20600, +51000, −20600) = (−30400, +51000, −20600). C crosses from None to Some on first touch.

7 Settlement under intraday trading (day 2): the anchor

The day-2 settle at $S_2 = 101$ runs the same fan-out of \$5 over holders A, B, C:

$$\text{VM}(A) = 6 \cdot 101 \cdot 50 + (-30400) = 30300 - 30400 = -100,$$

$$\text{VM}(B) = -10 \cdot 101 \cdot 50 + 51000 = -50500 + 51000 = +500, \quad \text{VM}(C) = 4 \cdot 101 \cdot 50 + (-20600) = 20200 - 20600 = -400$$

Conservation: $\sum \Delta \text{net_qty} = 0$ (no quantity moved), $\sum \text{VM} = -100 + 500 - 400 = 0$. The *ac* reset: $\text{ac} \leftarrow (-30300, +50500, -20200)$ from $\text{ac} \leftarrow -\text{net_qty} \cdot 101 \cdot 50$, and $\sum \Delta \text{ac} = 0$. State: *net_qty* = (6, −10, 4), *ac* = (−30300, +50500, −20200).

The load-bearing subtlety. A's day-2 variation margin is -100 , not the naive $\text{net_qty} \cdot (S_2 - S_1) \cdot m = 6 \cdot (101 - 102) \cdot 50 = -300$. A sold 4 contracts at 103 intraday, one point above the prior mark 102, gaining $4 \cdot (+1) \cdot 50 = +200$, which offsets the -300 mark loss on the held position. Only the stored per-wallet **ac**, which absorbed the intraday trade, yields the correct -100 . A price-derived model gives the wrong cash. This is why **ac** is per-position stored state (C11), not a consequence read off the shared price.

The three answers, stated plainly.

1. *Settlement is a state update; its parts split by layer.* The shared part — **last_settlement_price**, **last_settlement_date** — is one write on *UnitStatus* $[u_F]$, one value per contract. The per-position part — the **ac** reset and the cash leg — is a fan-out over the current holders on *PositionState* $[w, u_F]$. The event touches both layers in one atomic **StateDelta**.
2. *It is one atomic event that fans out, not a derived consequence of the price.* The cash leg is real daily money moving between longs and shorts through CH; every cash move is a recorded, conservation-bearing event, so a per-holder pass is unavoidable. The **ac** reset rides in the same atomic delta under the single-writer discipline. The fan-out is forced by the cash, not chosen for the bookkeeping.
3. *Price only in shared state, consequence only in per-wallet state.* **last_settlement_price** lives only in *UnitStatus* $[u_F]$; its consequence — the **ac** reset and the variation-margin cash — lives only in *PositionState* $[w, u_F]$ and the move stream. The price is shared; its consequence is per-wallet.

8 Closing a position to flat

B buys 4 contracts from C at 101; C goes flat. One **StateDelta**:

$$\Delta \text{net_qty}(B) = +4, \Delta \text{net_qty}(C) = -4; \quad \Delta \text{ac}(B) = -(+4) \cdot 101 \cdot 50 = -20200, \Delta \text{ac}(C) = +20200.$$

Conservation: $\sum \Delta \text{net_qty} = 0$, $\sum \Delta \text{ac} = 0$, $\sum \text{VM} = 0$ (no cash leg). State: $\text{net_qty} = (6, -6, 0)$, $\text{ac} = (-30300, +30300, 0)$. C is now flat. Its row is *retained*, Some at $(\text{net_qty } 0, \text{ac } 0)$ — not deleted, not None. The two readings are distinct and both load-bearing: None means never held; Some-flat means held and now flat, read after close-out for tax reporting, wash-sale lookback, and trade reconstruction (C1(a)). The carrier is monotone: once a row is created it is never removed (C1(b)); no row deleter exists, because the ledger is abstract.

9 Expiry and final settlement

Expiry is the settlement fan-out of §5 composed with a stage write to **EXPIRED**; like a settle, it acts only on an **ACTIVE** unit, never a never-traded one (§11). The final mark $S = 105$ fans out over the remaining holders A and B (C is flat, its row touched to no effect):

$$\text{VM}(A) = 6 \cdot 105 \cdot 50 + (-30300) = 31500 - 30300 = +1200, \quad \text{VM}(B) = -6 \cdot 105 \cdot 50 + 30300 = -1200.$$

Conservation: $\sum \Delta \text{net_qty} = 0$ (no quantity moved), $\sum \text{VM} = +1200 - 1200 = 0$; **ac** $\leftarrow (-31500, +31500, 0)$ with $\sum \Delta \text{ac} = 0$. The coarse rank advances to **EXPIRED**, the stage value becoming **Expired** (**Settlement 105 (Day 3)**); the final mark is always present.

Cash settlement and the Close. No unit is delivered; the economic result is the cumulative variation margin already moved at the final settle. The *Close* step extinguishes every holder's position against the clearinghouse, flattening both `net_qty` and `ac` by an additive negation of each row, with no further cash. A held +6 and B held -6; closing them returns the contracts to CH:

$$\Delta\text{net_qty}(A) = -6, \quad \Delta\text{net_qty}(B) = +6, \quad \Delta\text{net_qty}(\text{CH}) = 0; \quad \Delta\text{ac}(A) = +31500, \quad \Delta\text{ac}(B) = -31500,$$

The clearinghouse legs are the residual of the holder legs — zero, since the holder legs already balance — so no CH row materialises. Conservation, on all three sums: $\sum \Delta\text{net_qty} = -6 + 6 + 0 = 0$, $\sum \Delta\text{ac} = +31500 - 31500 + 0 = 0$, $\sum \text{VM} = 0$ (Close moves no cash). State after Close: `net_qty` = (0, 0, 0), `ac` = (0, 0, 0); rows are retained at zero (monotone carrier). Close carries no stage write, so it is the one event still admissible on an **EXPIRED** unit.

Physical settlement. The variant differs only at the Close. Instead of flattening against CH for cash, the contract converts to delivery: the deliverable unit and the cash payment move together as one atomic transaction (delivery versus payment), the future's `net_qty` goes to zero, and the holder's wallet gains the deliverable. The cash leg is struck at the *final settlement price* S , not the trade price: expiry has already marked every position to S through variation margin, so delivery exchanges the deliverable for $\text{net_qty} \cdot S \cdot m$ and moves no further mark-to-market — pricing the delivery at any other figure would double-count the variation margin already settled. Per-unit conservation holds on each unit separately — the future, the deliverable, the cash — and the cross-unit delivery is one transaction applied all-or-nothing. The worked example is cash-settled; the figures above are unchanged by this variant.

Closing identity. Cumulative variation margin per wallet: $A = +1000 - 100 + 1200 = +2100$; $B = -1000 + 500 - 1200 = -1700$; $C = -400$; CH = 0; sum 0. Each equals the wallet's economic profit and loss: $A = 4 \cdot (103 - 100) \cdot 50 + 6 \cdot (105 - 100) \cdot 50 = +2100$; $B = 4 \cdot (100 - 101) \cdot 50 + 6 \cdot (100 - 105) \cdot 50 = -1700$; $C = 4 \cdot (101 - 103) \cdot 50 = -400$. Cash ties to economic profit and loss by construction.

10 After expiry: state versus derivation

After the Close, every row is retained at zero: $\text{PositionState}[w, u_F] = \text{Some flat}$ for A, B, C (monotone carrier). $\text{UnitStatus}[u_F]$ holds stage **EXPIRED** and final mark 105. $\text{ProductTerms}[u_F]$ is unchanged — immutable.

`first_touch_date` is *not* stored: it is derived from the event log on demand. Caching it in PositionState would make a replay disagree with itself under a back-dated correction, the cached value reflecting the pre-correction order while the fold reflects the corrected one. The rule is general: what the fold over the log determines is derived, not stored; only what the fold cannot reconstruct from prior events is state.

11 Invariants threaded through the life

- **Conservation per event (C2).** Every `StateDelta` satisfies $\sum_w \Delta\text{net_qty} = 0$, $\sum_w \Delta\text{ac} = 0$, $\sum_w \text{VM} = 0$, discharged by `validate` before application. Induction over the stream lifts the per-event identity to the global invariant. The vacuous case — a unit with no holders — is the empty sum, zero (C9); conservation sums deltas and never divides by a holder count.
- **Deterministic replay as a fold (C1(b)).** `replay` is a Kleisli fold of the event stream over the ledger; $\text{replay}(xs ++ ys) = \text{replay } xs \gg \text{replay } ys$. Checkpoint independence is a consequence of this law, not a test; the monotone carrier keeps the key set stable across cuts. The stream is assumed *unique* — de-duplicated at the ingestion boundary by an event key —

because the two event kinds differ in idempotency: a duplicated `SettleVM` at a fixed mark is inert ($\Delta ac = 0$, $VM = 0$, by the no-op above), whereas a duplicated `Trade` accumulates a second position delta. Duplicate suppression is therefore a boundary obligation, not a ledger function.

- **ac per-position (C11).** Accumulated cost is keyed by (w, u) with a single canonical writer (settle/trade). This is what makes the intraday trader’s variation margin correct (§7).
- **Settlement price shared.** `last_settlement_price` is one value per contract on `UnitStatus[uF]`, read identically by every holder.
- **Monotone, absorbing transitions.** The coarse stage rank `REGISTERED < ACTIVE < EXPIRED` never regresses; a proposed stage of strictly lower rank is rejected (`StageRegression`). Settle and expiry act only on an `ACTIVE` unit; both reject a `REGISTERED` (never-traded) unit (`NotActive`), which has no holders and no mark slot to update — so a never-traded unit is never silently promoted, and `EXPIRED` is reachable only from `ACTIVE`: the chain is `REGISTERED→ACTIVE→EXPIRED` with no skips. `EXPIRED` is additionally *absorbing*: no trade, settle, or re-expiry acts on an expired unit. The rank guard alone is too weak here — a second expiry has equal rank, and $2 < 2$ is false — so an explicit absorbing test rejects every stage-writing delta on an expired unit; `Close`, which carries no stage write, is the one event still admissible. Re-settling at the same price before expiry is a no-op: with $ac = -net_qty \cdot S \cdot m$ already, $\Delta ac = 0$ and $VM = 0$ — the transition is idempotent at a fixed mark.

12 Escalations

E1 — Fan-out cost at scale. A settlement writes one `PositionState` row and emits one cash leg per open holder, every day. Across $\sim 10^6$ contracts each with up to thousands of holders, daily settlement is $O(\text{open positions})$ writes and cash legs — a large daily fan-out and write amplification, the same key-space pressure as addendum risk F3. This is intrinsic to per-wallet variation margin. Mitigations (batching the fan-out, snapshotting) trade against the conservation-per-event and single-writer disciplines and must be shown to preserve C2 and C11. The cost stands. *Owner: the performance/risk agent.*

E2 — The derived-consequence alternative, declined. Storing only the shared `last_settlement_price` and computing each wallet’s mark lazily saves nothing on the dominant cost — the cash leg is per-wallet and unavoidable (E1) — cannot produce the correct variation margin for an intraday trader without reconstructing `ac` anyway, and would split the single-writer discipline for `ac` (C11). The fan-out is forced by the cash, not chosen for the bookkeeping.